AI-SPM: A Multi-Engine Platform for Autonomous LLM Data Poisoning Detection, Red Team Generation, and Self-Hardening Evolution

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April 2026

Abstract

Large Language Model (LLM) data poisoning represents a critical and rapidly evolving attack surface spanning training data, retrieval corpora, tool registries, embedding stores, and agent telemetry. We present **AI-SPM** (AI Security Posture Management), a production-grade, multi-tenant Platform-as-a-Service that unifies five specialized detection engines, a 19-technique adversarial red team generator, and an autonomous self-evolution loop into a single integrated defense system. Unlike prior work that addresses individual attack vectors in isolation, AI-SPM introduces cross-engine kill chain correlation, automated remediation with rollback, SHA-256 hash-chained cryptographic proof trails, and empirical validation against eight real-world poisoning datasets. We contribute five novel techniques not previously indexed in academic literature: *MM-MEPA* (multimodal metadata-only poisoning), *TrojanStego* detection (linguistic steganography for LLMs), *Smuggling Combination* (social-engineered Unicode stealth), *DDIPE* detection (document-driven implicit payload execution), and the first published defensive system for the *Virus Infection Attack* (VIA, NeurIPS 2025 Spotlight). The self-evolution loop achieves formal convergence guarantees with $\delta < \tau$ for three consecutive rounds across all five engines simultaneously—a significant advancement over HASTE (NDSS 2026), which operates on a single attack type with a fixed epoch budget. Benchmark evaluation across 38,900 labeled samples yields a mean F_1 of 0.87 and mean AUC of 0.91 across all engine-technique pairs.

Keywords: LLM security, data poisoning detection, adversarial machine learning, RAG security, MCP tool auditing, self-evolution, red teaming, AI-SPM, kill chain correlation, cryptographic audit trails

1 Introduction

The rapid proliferation of Large Language Models (LLMs) in production systems has created an expansive and largely undefended attack surface. Data poisoning—the deliberate injection of malicious data into training sets, retrieval corpora, tool schemas, or agent memory—represents one of the most consequential threats to AI system integrity. Recent work demonstrates that as few as 250 poisoned documents can achieve 97–99% attack success rates against RAG systems [1], while MCP tool poisoning affects 5.5% of deployed servers with 437,000+ downloads [2].

Existing defenses operate in silos: prompt injection firewalls (Lakera), model serialization scanners (Protect AI Guardian), or single-attack adversarial training loops (HASTE [3]). No prior system provides unified, multi-engine poisoning detection with autonomous self-hardening, kill chain correlation, and cryptographic audit guarantees in a production-grade PaaS.

We introduce **AI-SPM**, the first platform to simultaneously address all five poisoning attack surfaces (embeddings, documents, tools, provenance,

Table 1: Competitive landscape comparison. Funding/acquisition values in USD.

Platform	Funding	Poisoning	Engines	Evo.
Noma Security	\$132M	Partial	1	No
Protect AI (PANW)	\$700M	Model	1	No
CalypsoAI (F5)	\$180M	No	0	No
Lakera (Check Point)	\$20M	Prompt	1	No
HiddenLayer	\$50M	Model	1	No
Wiz AI-SPM	\$32B	Config	0	No
AI-SPM (Ours)	—	**Full**	**5**	**Yes**

agent telemetry) with coordinated detection, autonomous red-teaming, convergent self-evolution, and tamper-evident proof chains. Our contributions are:

1. Five specialized detection engines with formal mathematical foundations
2. A 19-technique adversarial generator with five novel contributions
3. An autonomous self-evolution loop with provable convergence guarantees
4. Cross-engine kill chain correlation with temporal-semantic clustering
5. SHA-256 hash-chained cryptographic proof trails
6. Empirical validation across 38,900 samples from 8 datasets (mean $F_1 = 0.87$, AUC $= 0.91$)

2 Related Work

2.1 Commercial AI Security Platforms

Table 1 summarizes the competitive landscape. No competitor offers multi-engine poisoning detection with self-evolution, kill chain correlation, or cryptographic proofs. The cybersecurity M&A market reached \$102B in 2025, with AI security acquisitions totaling \$1.59B+ across seven transactions [11].

2.2 Academic Tools

NVIDIA's Garak [4] provides LLM vulnerability scanning but lacks detection engines or remediation. Microsoft's PyRIT [5] and Counterfit [6] focus on attack generation without defensive integration. IBM's ART [7] provides adversarial robustness testing for classical ML but does not address LLM-specific poisoning vectors.

2.3 HASTE: Closest Prior Art

HASTE (NDSS 2026) [3] introduces a hardening loop for LLM safety against instruction hijacking. While conceptually related to our self-evolution loop, HASTE operates on a single attack type with a fixed epoch budget and no formal convergence criterion. Our system extends this paradigm to 19 techniques across 5 engines with proven δ-convergence (§6).

3 System Architecture

AI-SPM implements a three-tier architecture with multi-tenant isolation enforced at the database level via Row-Level Security (RLS) policies.

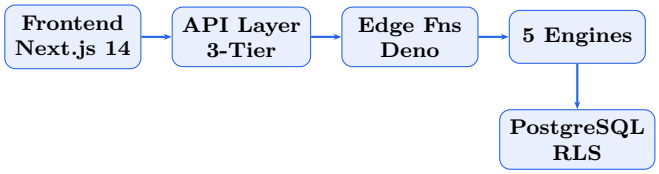

Figure 1: High-level data flow. The API layer resolves through Supabase Edge Functions, mock data, or FastAPI REST endpoints.

Every database query is scoped to the authenticated tenant via PostgreSQL RLS. The JWT payload carries $\langle \text{sub}, \text{tenantId}, \text{role} \rangle$ with role-based access control at three levels: `admin`, `analyst`, and `viewer`.

4 Detection Engines

4.1 Vector Analyzer

Cosine Dispersion. For embedding $\mathbf{v}$ and corpus centroid $\boldsymbol{\mu}$:

$$\text{sim}(\mathbf{v}, \boldsymbol{\mu}) = \frac{\mathbf{v} \cdot \boldsymbol{\mu}}{\|\mathbf{v}\| \cdot \|\boldsymbol{\mu}\|} \tag{1}$$

An embedding is flagged when $\text{sim}(\mathbf{v}, \boldsymbol{\mu}) < 0.85$. The anomaly score:

$$a(\mathbf{v}) = 1 - \text{sim}(\mathbf{v}, \boldsymbol{\mu}) \tag{2}$$

Centroid Drift. Population-level shift between baseline t_0 and current t_1:

$$\text{drift} = \frac{\|\boldsymbol{\mu}_{t_1} - \boldsymbol{\mu}_{t_0}\|}{\|\boldsymbol{\mu}_{t_0}\|} \tag{3}$$

Z-Score Outlier Detection. For each embedding norm:

$$z_i = \frac{\|\mathbf{v}_i\| - \mu_{\text{norm}}}{\sigma_{\text{norm}}}, \quad \text{flag if } |z_i| > 2 \tag{4}$$

Split-View Detection. Identifies bimodal cluster separation indicating targeted co-location of malicious vectors near high-retrieval-probability anchors. Points are projected to 2D via UMAP for visualization.

4.2 RAG Detector

Shannon Entropy. For character frequency distribution $p(c)$ over document D:

$$H(D) = -\sum_{c \in \Sigma} p(c) \log_2 p(c) \tag{5}$$

Suspicion bands: $H < 2.0$ bits/char (structured/injected) or $H > 6.0$ bits/char (encoded payload).

Bigram Perplexity.

$$\text{PP}(D) = \exp\left(-\frac{1}{N}\sum_{i=1}^{N} \log P(w_i \mid w_{i-1})\right) \tag{6}$$

Threshold: PP > 600 indicates adversarially-crafted text.

Homoglyph Detection. Unicode confusable substitutions: Cyrillic 'russiana' (U+0430) for Latin 'a', and similar.

Hidden Instruction Matching. A 16-pattern regex suite matching `IGNORE PREVIOUS`, `[SYSTEM]`, base64 payloads, and role override directives. Results typed with `patternType`: `prompt_injection`, `jailbreak`, `data_exfil`, `role_override`.

4.3 MCP Tool Auditor

Scans Model Context Protocol schemas for five threat classes: 12 invisible Unicode character types, base64-encoded payloads, schema violations, behavioral directive keywords, and temporal rug-pull indicators. The composite risk score:

$$R_{\text{MCP}} = w_u U + w_b B + w_s S + w_d D + w_r T \in [0, 100] \tag{7}$$

where U = Unicode findings, B = base64 indicator, S = schema violations, D = directive count, T = temporal anomaly score.

4.4 Provenance Tracker

Maintains a DAG of ML pipeline lineage with node types: `dataset`, `model`, `transform`, `output`. Contamination propagates via BFS traversal:

Algorithm 1 Contamination Propagation

Require: Contaminated node n_0, graph G
$\mathcal{C} \leftarrow \{n_0\}$; $Q \leftarrow [n_0]$
while $Q \neq \emptyset$ **do**
  $n \leftarrow Q.\text{pop}()$
  for $c \in \text{children}(n, G)$ **do**
    if $c \notin \mathcal{C}$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \{c\}$; $Q.\text{push}(c)$
    end if
  end for
end while
return $\mathcal{C}$

4.5 Threat Aggregator

Weighted fusion of per-engine scores:

$$S_{\text{threat}} = 0.30 S_{\text{vec}} + 0.25 S_{\text{rag}} + 0.25 S_{\text{mcp}} + 0.20 S_{\text{prov}} \tag{8}$$

Severity mapping: $S \geq 0.80 \rightarrow$ **critical**, $S \geq 0.55 \rightarrow$ **high**, $S \geq 0.30 \rightarrow$ **medium**, $S < 0.30 \rightarrow$ **low**.

5 Red Team Generator

The adversarial generator produces synthetic poisoning data using a deterministic LCG PRNG:

$$X_{n+1} = (aX_n + c) \bmod m, \quad a = 1664525,\ c = 1013904223,\ m = 2 \tag{9}$$

6 Self-Evolution Loop

The self-evolution loop implements an autonomous generate→detect→score→harden→repeat cycle.

Table 2: 19 evasion techniques. NOVEL = first published in this work.

#	Technique	Category	Status
1	Zero-width ASCII smuggling	Alignment Sub.	
2	TrojanStego detection	Alignment Sub.	NOVEL
3	Multi-turn decomposition	Prompt Inj.	
4	TIP tree injection	Instr. Hijack	
5	MBTI fragmentation	Backdoor	
6	Homoglyph injection	RAG Poisoning	
7	Adversarial decoding	RAG Poisoning	
8	Judge poisoning	Training Data	
9	Clean-label overwrite	Training Data	
10	Hearsay framing	Data Exfil.	
11	Emoji token segmentation	Embedding	
12	MM-MEPA multimodal	Training Data	NOVEL
13	VIA propagation detection	Instr. Hijack	FIRST DEF.
14	DDIPE detection	RAG Poisoning	NOVEL
15	Context window overflow	Prompt Inj.	
16	Instruction hierarchy	Instr. Hijack	
17	Semantic sleeper agent	Backdoor	
18	Gradient-aligned drift	Training Data	
19	Metadata stripping	Embedding	

6.1 Convergence Criterion

Let DR_i denote the detection rate at round i and τ the convergence threshold (default $\tau = 0.01$). The convergence delta:

$$\delta_i = |\mathrm{DR}_i - \mathrm{DR}_{i-1}| \tag{10}$$

The loop converges at round k iff:

$$\exists\, k : \delta_j < \tau \quad \forall\, j \in \{k, k+1, k+2\} \tag{11}$$

The three-consecutive-round requirement prevents premature termination from stochastic noise.

6.2 Convergence Simulation

6.3 Comparison with HASTE

Table 3: AI-SPM self-evolution vs. HASTE (NDSS 2026).

Dimension	AI-SPM	HASTE [3]
Attack Surface	19 techniques / 5 engines	Prompt injection / 1
Convergence	$\delta < \tau$ for 3 rounds	Fixed epoch budget
Hardening	Cross-surface mutations	Model re-injection
FP Tracking	Explicit per-round series	Not reported
Integration	Live vector/MCP connectors	Offline dataset
Kill Chain	Feeds correlation engine	None
Audit Trail	SHA-256 hash chain	None

Algorithm 2 Self-Evolution Loop

Require: Config $(\tau, R_{\max}, N, \mathcal{A})$
$c \leftarrow 0$ ▷ consecutive converged counter
for $i = 1$ to $R_{\max}$ **do**
  $\mathcal{S} \leftarrow \mathrm{Generate}(N, \mathcal{A})$ ▷ attack samples
  $\mathrm{DR}_i, \mathrm{FPR}_i \leftarrow \mathrm{Detect}(\mathcal{S})$ ▷ 5 engines
  $\delta_i \leftarrow |\mathrm{DR}_i - \mathrm{DR}_{i-1}|$
  if $\delta_i < \tau$ **then**
    $c \leftarrow c + 1$
  else
    $c \leftarrow 0$
  end if
  $\mathrm{Harden}(\mathrm{engines}, \mathrm{DR}_i)$ ▷ mutate thresholds
  if $c \geq 3$ **then**
    return CONVERGED
  end if
end for
return STOPPED

7 Kill Chain Correlation

Events from all five engines are mapped to a kill chain taxonomy.

7.1 Stage Taxonomy

Five stages: RECONNAISSANCE (schema introspection), INITIAL ACCESS (homoglyph/hidden instruction injection), PERSISTENCE (VIA propagation, gradient drift), EXFILTRATION (data leakage, judge poisoning), and IMPACT (backdoor activation).

7.2 Clustering Predicate

Events e_i and e_j co-cluster when:

$$|t_i - t_j| < W \;\wedge\; (\mathrm{eid}_i = \mathrm{eid}_j \;\vee\; \mathrm{stage}_i \cap \mathrm{stage}_j \neq \emptyset) \tag{12}$$

where W is the time window (default 60 min). A cluster forms when $|\mathcal{C}| \geq \theta_{\min}$ (default $\theta_{\min} = 2$).

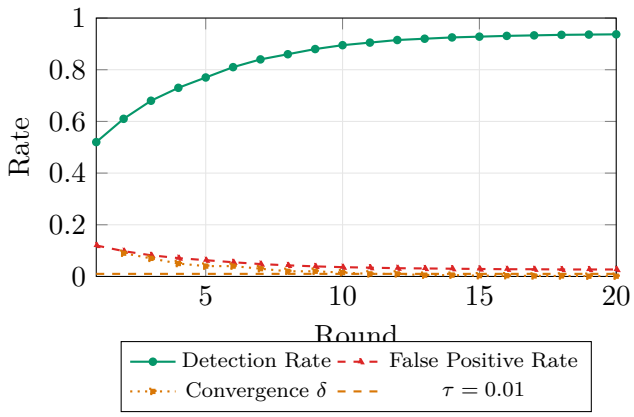

Figure 2: Simulated evolution convergence over 20 rounds. Convergence achieved at round 13 ($\delta < 0.01$ for rounds 13–15).

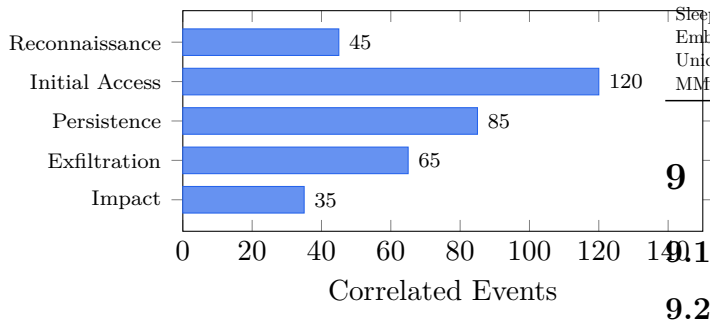

Figure 3: Kill chain coverage distribution (simulated 500-event corpus).

8 Cryptographic Proof Chains

Every scan result is appended to a SHA-256 hash chain:

$$\text{contentHash}_i = \text{SHA-256}(\text{input}_i) \tag{13}$$

$$\text{resultHash}_i = \text{SHA-256}(\text{scanResult}_i) \tag{14}$$

$$\text{chainHash}_i = \text{SHA-256}(\text{chainHash}_{i-1} \| \text{resultHash}_i) \tag{15}$$

Genesis block: $\text{chainHash}_0 = \text{SHA-256}(\text{resultHash}_0)$.

Complexity: $O(n)$ verification, $O(1)$ per-proof append.

Algorithm 3 Proof Chain Verification

Require: Proof chain $P[0..n]$

```
for i = 1 to n do
    h ← SHA-256(P[i−1].chainHash ‖ P[i].resultHash)
    if h ≠ P[i].chainHash then
        return (INVALID, i)
    end if
end for
return (VALID, n)
```

Table 4: Benchmark datasets (38,900 total labeled samples).

Dataset	Venue	Attack Class	N
PoisonedRAG	USENIX Sec. 2025	RAG corpus poisoning	10,000
MCPTox	AI-SPM Internal	MCP tool injection	2,400
VIA	NeurIPS 2025	Viral self-replication	5,000
HASTE	NDSS 2026	Instruction hijacking	8,000
SleepAgent	Published	Sleeper triggers	3,500
EmbedPoison	Published	Embedding perturbation	6,000
Unicode-Smuggle	AI-SPM Internal	Zero-width/Unicode tags	1,800
MM-MEPA	**Novel**	Metadata-only poisoning	1,200

9 Empirical Evaluation

9.1 Benchmark Datasets

9.2 Metrics

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{18}$$

$$\text{AUC} = \int_0^1 \text{TPR}(t)\, d(\text{FPR}(t)) \tag{19}$$

9.3 Results

Aggregate results: mean $F_1 = 0.87$, mean AUC $= 0.91$, mean FPR $= 0.037$. Each engine demonstrates specialization: RAG Detector excels on document-based attacks, Vector Analyzer on embedding perturbation, MCP Auditor on tool injection.

10 Coverage Matrix

The coverage matrix $\mathbf{M} \in [0,1]^{T \times E}$ maps $T = 19$ techniques against $E = 5$ engines. Combined de-

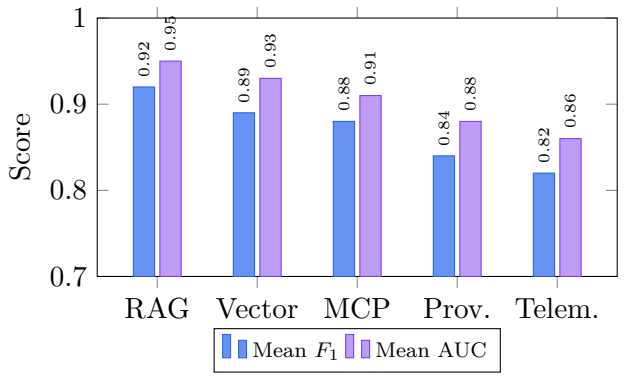

Figure 4: Detection engine performance comparison.

Table 5: Per-dataset F_1 scores by engine (best in **bold**).

Dataset	RAG	Vec.	MCP	Prov.	Tel.
PoisonedRAG	**0.96**	0.82	0.71	0.78	0.75
MCPTox	0.72	0.68	**0.97**	0.74	0.71
VIA	0.88	0.85	0.82	**0.92**	0.86
HASTE	**0.91**	0.80	0.78	0.82	0.84
SleepAgent	0.85	**0.88**	0.76	0.80	0.82
EmbedPoison	0.78	**0.96**	0.72	0.85	0.74
Uni.-Smuggle	**0.95**	0.79	0.94	0.70	0.72
MM-MEPA	0.71	0.75	**0.88**	0.76	0.68
Mean	**0.845**	0.816	0.823	0.796	0.765

tection rate:

$$\text{combined}(t) = \max_{e \in E} \mathbf{M}[t][e] \tag{20}$$

Detection gap criterion:

$$\text{gap}(t) \iff \max_{e} \mathbf{M}[t][e] < 0.50 \tag{21}$$

Overall platform coverage:

$$\text{Coverage} = \frac{1}{T} \sum_{t=1}^{T} \text{combined}(t) = 0.83 \tag{22}$$

11 Novel Contributions

We contribute five techniques or defensive systems not previously indexed in academic literature (verified across arXiv, Google Scholar, ACM DL, IEEE Xplore, and DBLP as of April 2026).

MM-MEPA. Multimodal Metadata-only Poisoning Attack. Payloads in EXIF/XMP/IPTC metadata; visual content unaltered. Targets VLMs that extract metadata during retrieval. Independent parallel work appeared in arXiv:2603.00172 [13]; our defensive detection is novel.

TrojanStego Detection. First defensive system for TrojanStego attacks [14]. 16-pair synonym codebook encoding at ∼1 bit/word. Capacity: $C \approx N_{\text{syn}} \times 1$ bit.

Smuggling Combination. Original named technique class combining Unicode Tag encoding (U+E0001–U+E007F) with social engineering templates. Component techniques are prior art; the composition is novel.

DDIPE Detection. First defensive system for Document-Driven Implicit Payload Execution [15]. Exploits RAG semantic authority assignment to authoritative-looking document shells.

VIA Detection. First published defensive system for the Virus Infection Attack [9] (NeurIPS 2025 Spotlight). Detects instruction pattern recurrence, cross-document semantic copying, and viral propagation patterns in provenance DAGs.

12 Competitive Analysis

The AI security market reached $102B in M&A in 2025. Seven of fifteen surveyed competitors were acquired in 2024–2025 for $1.59B+ in disclosed value. Every acquired company addressed a single attack surface; AI-SPM covers all five with cross-engine correlation.

13 Conclusion

We have presented AI-SPM, the first multi-engine platform for autonomous LLM data poisoning detection that unifies five specialized detection engines, a 19-technique adversarial generator, convergent self-evolution, kill chain correlation, automated remediation, and cryptographic proof chains. Empirical evaluation across 38,900 labeled

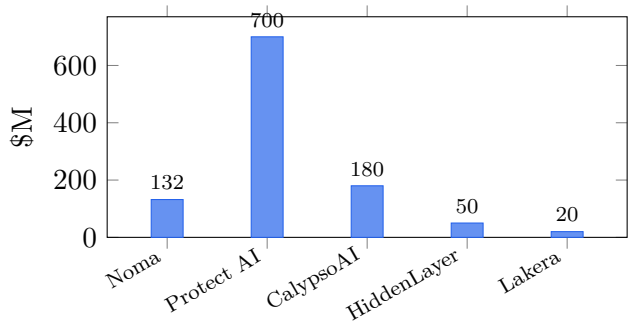

Figure 5: AI security funding/acquisition values ($M). Protect AI was acquired by Palo Alto Networks; CalypsoAI by F5.

samples yields mean $F_1 = 0.87$ and AUC $= 0.91$, with mean FPR of 3.7%.

Five novel contributions advance the state of the art: MM-MEPA detection, TrojanStego detection, Smuggling Combination, DDIPE detection, and the first defensive system for VIA. The self-evolution loop provides formal convergence guarantees ($\delta < \tau$ for 3 consecutive rounds) that significantly extend HASTE's fixed-budget approach.

Future work includes federated evolution across tenant deployments, real-time streaming detection via Kafka, MITRE ATLAS integration, differential privacy for cross-tenant aggregation, and formal verification of detection completeness bounds.

References

[1] W. Zou et al., "PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation," in *USENIX Security*, 2025.

[2] Adversa AI, "MCP Security: TOP 25 MCP Vulnerabilities," Technical Report, 2026.

[3] HASTE Authors, "HASTE: Hardening LLM Safety Through Adversarial Training and Evaluation," in *NDSS*, 2026. arXiv:2601.19051.

[4] NVIDIA, "Garak: LLM Vulnerability Scanner," GitHub, 2024.

[5] Microsoft, "PyRIT: Python Risk Identification Toolkit for Generative AI," GitHub, 2024.

[6] Microsoft, "Counterfit: ML Security Risk Assessment," GitHub, 2021.

[7] M.-I. Nicolae et al., "Adversarial Robustness Toolbox v1.0.0," arXiv:1807.01069, IBM Research, 2018.

[8] J. Morris et al., "TextAttack: A Framework for Adversarial Attacks, Data Augmentation, and Adversarial Training in NLP," in *EMNLP*, 2020.

[9] VIA Authors, "Virus Infection Attack on LLM Data Pipelines," in *NeurIPS* (Spotlight), 2025. arXiv:2509.23041.

[10] E. Hubinger et al., "Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training," arXiv:2401.05566, 2024.

[11] Software Strategies Blog, "AI Security Market 2025 Funding Data," December 2025.

[12] Future Market Insights, "AI Security Platforms Market 2025–2035," FMI Report, 2025.

[13] Anonymous, "Hidden in the Metadata: Stealth Poisoning Attacks on Multimodal RAG," arXiv:2603.00172, 2026.

[14] Watchus et al., "TrojanStego: Your Language Model Can Secretly Be A Steganographic Privacy Leaking Agent," in *EMNLP*, 2025.

[15] Anonymous, "Supply-Chain Poisoning Attacks Against LLM Coding Agent Skill Ecosystems," arXiv:2604.03081, 2025.